Foundation of Embodied AI: Course Map

2026-10-21

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Revision	2026-10-21
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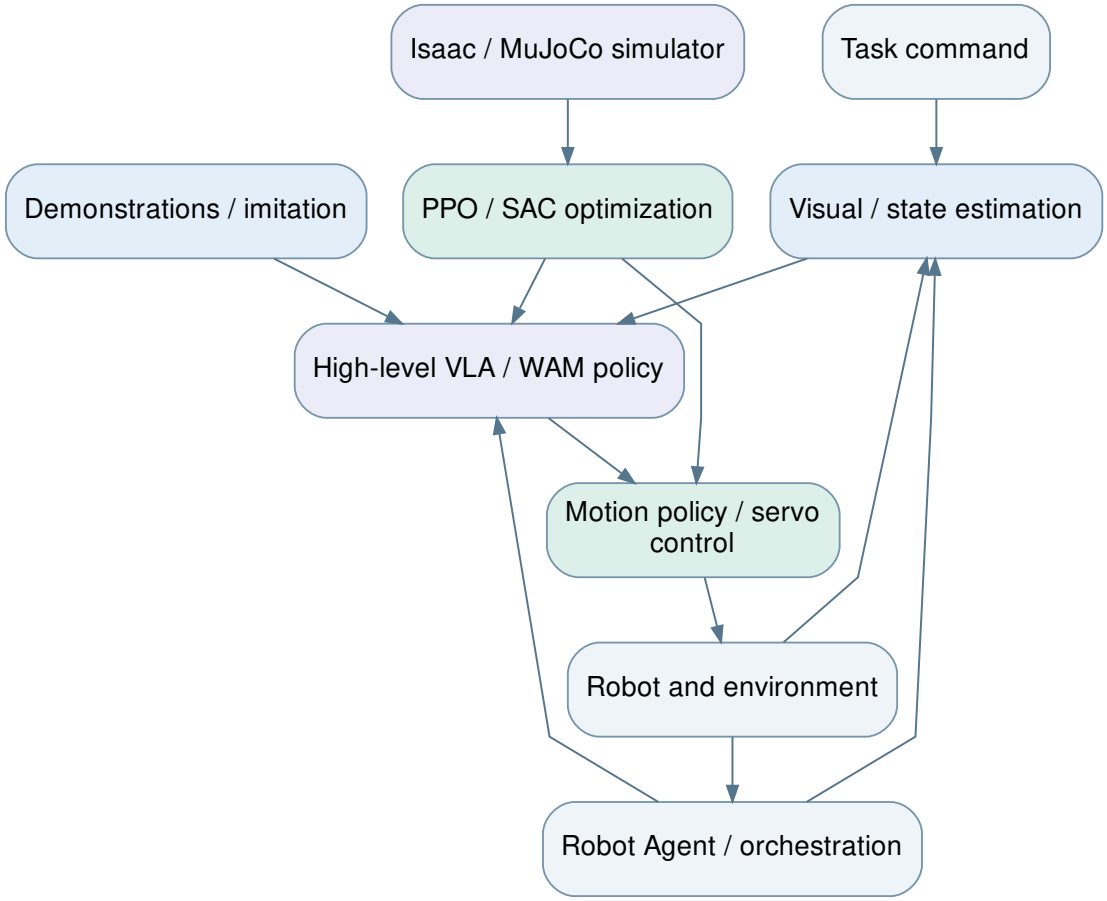
Six learning modules

Seventeen topics are organized as tools → calibration and deep learning → physics and perception → robot RL → demonstrations and generative policies → systems integration. Long-form derivations remain optional deep dives rather than strict prerequisites.

Module	Topics	Outcome
0. Orientation	Systems map and guide	Explain perception–policy–control loops
1. Tools	Git, ROS 2	Manage code and understand ROS communication and TF
2. Physics, perception and neural foundations	Calibration, Deep Learning Basics, vision foundation models, dynamics, Isaac / MuJoCo	Relate calibrated geometry, neural networks, physical models and simulation
3. Reinforcement learning	Motion RL, PPO, SAC	Design observations, actions, rewards and data collection
4. Demonstrations and models	Imitation Learning, Flow Matching, GR00T / SONIC, World Action Models, Efficient Training & Inference, Post-training	Learn generative and predictive policies and connect efficient inference with hierarchical controllers
5. Robot Agent integration	Robot Agent planning, tool use, verification and recovery	Orchestrate perception, policy and control for long-horizon tasks

Embodied AI system map



Reading paths

1. Beginners: orientation → essential Git and ROS 2 → calibration → deep learning → dynamics → vision foundation models → simulation.
2. Motion control: dynamics → simulation → deep learning → PPO / SAC → motion control RL.
3. VLA: calibration → deep learning → vision foundation models → Imitation Learning → Flow Matching → GR00T / SONIC → World Action Models → efficient training and inference → post-training → Robot Agent.
4. Mathematical deep dives: full PPO, SAC, and Flow Matching derivations are reference material; hands-on learning does not require deriving everything first.

Tutorial overview

1. Git Basics

Git Basics — Learn Git's working tree, staging area and commit history, then follow a GitHub workflow covering branches, pull requests, conflict resolution and common recovery commands.

2. ROS 2 Basics

ROS 2 Basics — Understand nodes, topics, services, actions, QoS and TF. Connect sensors, state estimation, policy inference and controllers through ROS 2 interfaces.

3. Camera Calibration

Camera Calibration — Study camera intrinsics, distortion, extrinsics and hand–eye calibration. Relate RGB-D observations and coordinate transforms to robot-frame object poses.

4. Deep Learning Basics

Deep Learning Basics — Learn supervised objectives, gradient descent and backpropagation, then connect MLPs, Attention and Transformers to multimodal robot action heads. Understand pretraining, fine-tuning and the foundations required by downstream policy methods.

5. Vision Foundation Models

Vision Foundation Models — Compare segmentation, pose estimation, stereo depth and 3D foundation models. Follow four robotics workflows spanning known objects, stereo geometry, unknown shapes and human demonstrations.

6. Robot Dynamics

Robot Dynamics — Develop rigid-body motion and dynamics through coordinate frames, Jacobians, inertia and equations of motion. Connect contact, friction and control models to robot simulation.

7. Isaac / MuJoCo Simulation

Isaac / MuJoCo Simulation — Understand Isaac Sim, Isaac Lab and MuJoCo as simulation and training tools. Cover asset import, environment contracts, RL library selection, parallel throughput, deployment interfaces.

8. Motion Control RL

Motion Control RL — Study motion tracking, DeepMimic/AMP, universal controllers, teacher–student learning, RMA and residual RL. Connect these methods to legged and dexterous motion, recovery and Sim-to-Real.

9. PPO / Actor–Critic

PPO / Actor–Critic — Derive PPO-Clip and Actor–Critic learning from policy gradients, Bellman targets and GAE. Explain Gaussian policy updates, gradient isolation and asymmetric robot observations.

10. Q-learning / SAC

Q-learning / SAC — Trace Q-learning, TD/SARSA and DQN toward entropy-regularized RL. Derive soft Bellman equations and modern SAC with twin critics, reparameterization and automatic entropy tuning.

11. Imitation Learning

Imitation Learning — Study BC, DAgger and corrective demonstrations, then connect ACT, Diffusion Policy and Flow Matching to action chunks. Explain VLA architectures and supervised action learning, temporal ensembling, and inference-time versus training-time RTC.

12. Flow Matching

Flow Matching — Derive conditional Flow Matching, velocity fields and the training objective. Explain how ODE integration transforms noise into conditional robot action sequences.

13. GR00T & SONIC

GR00T & SONIC — Use GR00T and SONIC to explain System 1/System 0 humanoid control. Connect high-level action chunks, motion representations, whole-body policies and low-level execution.

14. World Action Models

World Action Models — Distinguish world models, VLAs and WAMs, then study DreamZero training and accelerated inference, Dream-1 cross-embodiment deployment, and VLA/WAM memory, compute and latency trade-offs. Connect these methods to contact-rich control, recovery and evaluation.

15. Efficient Training & Inference

Efficient Training & Inference — Connect Transformer and VLA/WAM training costs, FlashAttention and valid caching to DreamZero acceleration, few-step inference, asynchronous robot control, RTC and cloud latency.

16. Post-training

Post-training — Compare SFT and RL for pretrained VLA and WAM models. Study six VLA RL routes, WAM-RL with online video SFT and Flow-SDE, then connect actor–critic semantics, RLinf integration and robot deployment limits.

17. Robot Agents / RPent

Robot Agents / RPent — Distinguish foundation models, coding agents and agent harnesses. Connect RAX exploration, ASPIRE trace repair and RPent toolkits, memory, Flash Mode and the RLinf data flywheel to reusable robot workflows.